Supplementary Materials

Covariance Estimation Matters for Multidimensional Normative Modeling: A Demonstration with Obsessive–Compulsive Disorder

Supplement S1: Matched-Floor Conditions

Two Study 1 conditions were added after Study 2 and are reported as exploratory, because they were designed to resemble one observed sample rather than to isolate a distributional property.

The first replaces the uniform floor proportion with the per-variable prevalences observed in the Study 2 control sample, where the percentage of controls at the scale minimum ranges from 22.6% to 93.5% across the six symptom measures. Each simulated symptom variable was censored at the prevalence of its corresponding measure, and the five personality variables carried no floor, as in the real sample. The second adds a latent correlation matrix derived from that sample. Because observed correlations are computed on data that already carry the floors, using them directly as the latent structure would attenuate the same effect twice; the observed correlations were therefore inverted through the censoring map and the result projected to the nearest positive-definite matrix.

The matched-floor vector behaved much as a uniform .50 floor did. Its benchmark calibration at $n = 62$ was a median ratio of 0.89 with 9.0% exceedance, against 0.84 with 10.7% for the uniform .50 floor, and the three estimators ordered as they did elsewhere, with the sample covariance at 0.91 with 7.6%, Ledoit–Wolf at 0.77 with 4.9%, and MCD at 1.48 with 42.6%. Ten of those 30 MCD fits were rank-deficient (7 of 30 with matched correlations), because the floor prevalence of the Snaith–Hamilton Pleasure Scale (.935) is higher than the .75 floor at which the uniform conditions first produced singular fits; their distances are formed with a

pseudoinverse. Treating each condition as a distribution across 30 seeds, all six quantities observed in the real control sample fell inside the 2.5th to 97.5th percentile range of this condition.

The second condition did not improve the correspondence. Imposing the reconstructed latent correlation matrix produced a narrower across-seed distribution that excluded the observed value on four of the six quantities, which is what imposing a structure estimated from 62 observations as though it were a population parameter would be expected to do.

Neither condition establishes that the simulated reference distribution describes the population from which these controls were drawn. Matching floor prevalence, and even matching an estimated correlation structure, constrains two features of a generating model that has many. The correspondence reported here is descriptive. The observed summaries fell within the simulated ranges for the matched-floor condition and outside those for the Gaussian and .10-floor conditions, which does not identify the generating model.

Supplement S2: Consequences for the Ordering of the Deviant Group

Estimation error in the reference covariance affects which observations a normative model ranks as most deviant, and that consequence is not proportional to the error in the covariance itself. Figure S1 gives Spearman rank agreement with the benchmark distances and top-decile Jaccard overlap, for the Gaussian and uniform-floor conditions at both reference sample sizes.

At a reference sample of 5,000 the two full-sample estimators reproduced the benchmark ordering at every floor level, both at a rank agreement of at least .998 with top-decile overlap between .91 and .93. MCD's rank agreement fell from .998 under Gaussian conditions to .883 at a .50 floor and .717 at a .75 floor, and its overlap from .932 to .575 and .377. At a reference

sample of 62 under a .50 floor, rank agreement was .911 for the sample covariance, .955 for Ledoit–Wolf, and .776 for MCD, with top-decile overlap of .554, .679, and .418. Ordering therefore degrades well before distances become uninterpretable, and it degrades most for the estimator whose covariance error does not resolve with sample size.

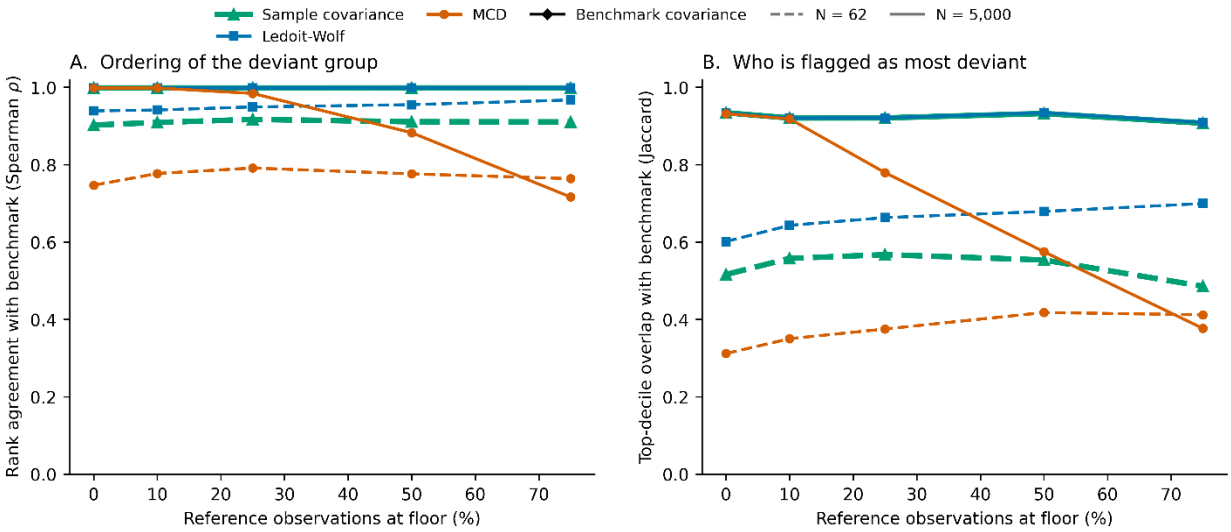


Figure S1. *Consequences of the reference covariance for the ordering of the deviant group in Study 1.* (A) Spearman rank agreement between each estimator's squared Mahalanobis distances and the benchmark distances. (B) Overlap between the top decile under each estimator and the top decile under the benchmark, as a Jaccard coefficient. Both are plotted against the proportion of the reference sample at a measurement floor, with the Gaussian condition at zero. Dashed lines give a reference sample size of 62 and solid lines 5,000. MCD = minimum covariance determinant.

Supplement S3: Calibration by Condition

Table S1 reports, for each condition at a reference sample of 5,000, the median squared Mahalanobis distance within the reference sample as a ratio of the chi-square median on eleven degrees of freedom, and the percentage of the reference sample beyond the chi-square .95 quantile. The benchmark columns give what these quantities become when the distances are formed with the mean and covariance the condition was generated from, and are the standard each estimator is read against. Main-text Table 4 gives the corresponding values at a reference sample of 62.

Two features of the table bear on interpretation. The benchmark itself departs from 1.00 and 5% wherever the marginals are non-normal, reaching 0.85 with 9.3% under skew and 0.70 with 14.0% under a .75 floor, so a chi-square cutoff misstates the tail of a floor-constrained reference distribution before any estimator is involved. And the contamination conditions must be read over the uncontaminated part of the reference group as well as the whole, because an estimator that correctly refuses to accommodate an outlier leaves that outlier far out, where it is then counted against the estimator.

Table S1

Calibration of Reference Distances by Condition at a Reference Sample of 5,000

Condition	Benchmark	Sample covariance	Ledoit–Wolf	MCD
Gaussian	1.00 / 4.9	1.00 / 4.9	1.00 / 4.8	1.00 / 4.9
Skew	0.85 / 9.3	0.85 / 9.3	0.85 / 9.2	1.19 / 22.7
Floor .10	0.99 / 5.0	0.99 / 5.0	0.99 / 5.0	1.00 / 5.3
Floor .25	0.97 / 7.1	0.97 / 7.1	0.97 / 7.1	1.12 / 14.2
Floor .50	0.88 / 11.6	0.88 / 11.6	0.88 / 11.6	2.33 / 63.4
Floor .75	0.70 / 14.0	0.70 / 13.9	0.70 / 13.9	—[a] / 99.4
Matched floors	0.88 / 8.0	0.88 / 8.0	0.87 / 7.9	1.97 / 53.6
Matched floors and correlations	0.89 / 7.8	0.89 / 7.8	0.88 / 7.7	2.76 / 70.9
Gaussian, 5% contaminated	1.03 / 8.7 (5.1)	0.98 / 6.0 (4.0)	0.98 / 5.9 (3.9)	1.02 / 8.0 (4.8)
Gaussian, 10% contaminated	1.08 / 12.0 (5.5)	0.99 / 5.7 (3.9)	0.99 / 5.6 (3.8)	1.05 / 9.6 (4.7)

Floor .50, 5% contaminated	0.94 / 14.9 (10.6)	0.87 / 11.4 (7.7)	0.87 / 11.3 (7.6)	2.58 / 86.3 (85.6)

Note. Values are across-seed medians for the median ratio and across-seed means for the percentage, at a reference sample of 5,000 over 30 random seeds. Each cell gives the median squared Mahalanobis distance as a ratio of the chi-square median on 11 degrees of freedom, followed by the percentage of the reference sample beyond the chi-square .95 quantile. For the contamination conditions the percentage is given over the whole reference group, with the percentage over its uncontaminated part in parentheses. MCD = minimum covariance determinant. At a .50 floor no fit was rank-deficient, but three of the thirty seeds returned a ratio above 10 and one above 10^8, which is what produces an across-seed mean of 9×10^6 beside a median of 2.33.

[a]At a .75 floor, 86.7% of MCD fits were rank-deficient at this reference sample size. Distances were formed with a pseudoinverse, so whether a near-null direction is inverted or discarded turns on a numerical cutoff, and the median ratio varies across seeds by orders of magnitude (across-seed SD = 2.5×10^{10}). No point value is reported. The proportion of rank-deficient fits is itself sensitive to the platform, moving by a seed or two between machines, so the main text gives it in words rather than to a decimal. The exceedance percentage is bounded and so does not explode in the same way, but it is computed from the same pseudoinverse distances and therefore describes the reduced-rank calculation rather than the intended eleven-dimensional one.

Supplement S4: Instrument Psychometrics

Reliability and validity evidence for each instrument from previously published samples. Internal consistency in the present sample is reported for every measure in Table 3 of the main text.

Beck Depression Inventory–II. Internal consistency is high, α = .91 (Dozois et al., 1998), and test–retest reliability is good, *r* = .73–.96 (Wang & Gorenstein, 2013).

Snaith–Hamilton Pleasure Scale. Internal consistency is high, α = .82 (Nakonezny et al., 2010), with stability over time, *r* = .70 (Franken et al., 2007).

Dimensional Obsessive–Compulsive Scale. Internal consistency is high, α = .90, and test–retest reliability is moderate, *r* = .66 (Abramowitz et al., 2010). Convergent validity is supported by correlations with other OCD measures, including the Obsessive–Compulsive Inventory–Revised and the Yale–Brown Obsessive Compulsive Scale (Foa et al., 2002; Goodman et al., 1989; Rapp et al., 2016).

Obsessive–Compulsive Trait Core Dimensions Questionnaire. Longer versions of this form demonstrate high internal consistency in both non-clinical (α = .89 for harm avoidance, α = .88 for incompleteness) and clinical samples, α = .92 and α = .91 (Summerfeldt et al., 2014), with good convergent validity with OCD severity (Lundström et al., 2024).

State–Trait Anxiety Inventory. Internal consistency is strong, α = .90 for state and α = .89 for trait, and adequate test–retest reliability (Barnes et al., 2002).

Rotterdam Symptom Checklist. A widely used measure of physical discomfort, with adequate reliability across physical-symptom subscales, α = .60 to .83 (Tchen et al., 2002), and convergent validity with measures of pain and discomfort, *r* = .50 (Pelayo-Alvarez et al., 2013).

Big Five Aspect Scales. Good internal consistency at the domain level (α = .80 to .89) and convergence with established Big Five inventories such as the NEO Personality Inventory–Revised, *r* = .80 to .92 (DeYoung et al., 2007). It has since been validated in further non-clinical adult samples and in Arabic, German, French, and Italian translations (Ali et al., 2024; DeYoung et al., 2016; Haehner et al., 2025).

Quality of Life Enjoyment and Satisfaction Questionnaire–Short Form. Good internal consistency (α = .90) and test–retest reliability, r = .93 (Stevanovic, 2011).

Supplement S5: Support Fraction Sweep

MCD's raw support size is user-specifiable. scikit-learn's default, $(n + p + 1)/2$, corresponds to 0.597 in the present sample and approaches 0.50 as n grows. One explanation for the estimator differences reported in Study 2 is therefore that the default is simply too aggressive, and that a less aggressive setting would yield a usable covariance. To test this, MCD was refit to the standardized healthy-control feature space at support fractions from 0.50 to 0.95 in steps of 0.05, with the full grid repeated across 30 random seeds. At each setting we recorded the size and symptom composition of the retained support set, the condition number of the fitted covariance, the Frobenius distance to a per-seed benchmark covariance constructed as in the Study 2 Method, whether the fitted matrix was invertible, and whether any feature was exactly constant within the retained subset. The same grid was applied to synthetic healthy-control samples of N = 2,000, generated by the bootstrap procedure described in the Study 2 Method, to establish whether any observed behavior depended on the size of the real control sample. Ledoit–Wolf and the untrimmed sample covariance are reported at every setting as reference points. Results appear in Table S2, Table S3, and Figure S2. Those ten settings across thirty seeds give the 300 fits that are the denominator for the singular-fit count reported below.

Two further quantities carry a support fraction of 1.00 and are not points on that grid. The untrimmed sample covariance and Ledoit–Wolf shrinkage discard no observations and have no support fraction, so they are reported at every setting as fixed reference lines rather than as MCD settings, and the sample covariance's Frobenius error should not be read as the endpoint of the sweep. MCD with its support fraction set to 1.00 was fitted across the same thirty seeds and

returned an identical fit in every one, since setting the fraction to 1.00 places every observation in the raw support set and leaves no subset to choose. The subsequent reweighting step can still exclude observations, so it is not the sample covariance.

MCD fits depend on the scikit-learn release. Under version 1.4.2 the default fit to these data returns a condition number of 155.71 and a reweighted support of 40, against 172.23 and 38 under version 1.8.0. All values reported here use 1.8.0.

Trimming removed the higher-symptom tail of the control distribution at every setting examined. The ratio of mean Dimensional Obsessive-Compulsive Scale (DOCS) scores in excluded versus retained control participants, averaged across seeds, was 7.46 (*SD* = 1.37) at a support fraction of 0.50, reached its maximum of 8.02 (SD = 1.09) at 0.65, and declined monotonically from there to 4.56 (*SD* = 0.17) at 0.95, so the mean ratio never fell below 4.5 (Table S2; Figure S2A). Ratios for individual seeds ranged from 3.94 to 11.24. The estimate was most precise where the ratio was smallest, so the conclusion that trimming never approaches unselective exclusion holds most firmly at the setting least likely to produce it. At 0.95, where only 8 of 62 control participants were discarded, those 8 still averaged more than four times the DOCS score of the 54 retained.

Loosening the trim reduced but did not remove the estimation bias. Frobenius distance to the benchmark covariance fell monotonically from 3.65 at 0.50 to 2.08 at 0.95, remaining well above both Ledoit–Wolf (1.12) and the untrimmed sample covariance fitted to the same 62 participants (0.05; Figure S2C). At the four support fractions where every seed produced an invertible matrix (0.65 through 0.80), condition numbers ran from 146.70 to 46.19 against a benchmark of 35.9 and an untrimmed value of 36.65. MCD at a support fraction of 1.00, identical across all thirty seeds, still excluded 5 of 62 controls at the reweighting step, whose

mean DOCS score was 6.6 times that of the retained controls. The resulting Frobenius error was 2.25, compared with 2.08 at a support fraction of 0.95, and the median OCD deviation was approximately 190, compared with 95 under the ordinary sample covariance. Including every observation in MCD's raw support therefore did not make its final reweighted estimate equivalent to the untrimmed sample covariance.

Nearly half of all fits produced no invertible matrix. Across the 300 MCD fits (10 support fractions × 30 seeds), 136 yielded a numerically singular covariance, distributed non-monotonically across the grid (Figure S2D, upper panel). Every one of these 136 fits, and none of the other 164, retained a support set on which the Snaith-Hamilton Pleasure Scale (SHAPS) was exactly constant. A feature with no variance within the retained subset forces a zero row and column in that subset's covariance, so singularity follows arithmetically. What the sweep establishes is that no other cause arose, and that a floor-constrained dimension collapsed to a constant in 45% of fits. This also accounts for the non-monotonic pattern. At low support fractions the retained subset is small enough that every retained participant may score zero, 31 of 62 at 0.50. At high fractions the reweighting step removes participants rather than adding them, 58 raw against 53.8 reweighted at 0.95 (Figure S2B), and those it removes are the ones furthest from the center, which are the participants scoring above floor. Where the fitted covariance was singular, the pseudoinverse used for scoring zeroed the SHAPS direction, so deviation in those fits was computed on ten dimensions rather than eleven, with no error raised.

The same sweep applied to synthetic healthy-control (HC) samples of $N = 2{,}000$ reproduced both the restriction ratios (4.89 to 9.32) and the Frobenius distances (2.23 to 3.73; Table S3), indicating that neither depends on the small size of the real control sample. Singular fits did not recur there, but the synthetic generator adds continuous jitter to every value and so

cannot produce the exact ties that drive degeneracy. That setting therefore speaks to bias and conditioning rather than to rank deficiency.

All analyses were conducted in Python using scikit-learn 1.8.0, with seeds 42 through 71. Numerical rank and every pseudoinverse come from scipy.linalg.pinvh at its default cutoff, and singularity is judged from the eigenvalue spectrum against machine precision scaled to the matrix rather than from a fixed threshold on the condition number. The script that produces Table S2, Table S3, and Figure S2 is available at https://doi.org/10.5281/zenodo.22878835, along with the package versions used.

Table S2

Real Healthy Control Sweep

Estimator	Support Fraction	Retained (Mean)	Restrict Ratio (Mean ± *SD*)	Condition No. (Mean)	% Singular	Frobenius Dist. (Mean ± *SD*)	Median D^2
MCD	0.50	31.6	7.46 ± 1.37	—	100.0[a]	3.65 ± 0.08	1,584.10
MCD	0.55	34.8	6.98 ± 1.62	154.82	66.7[a]	3.52 ± 0.10	1,464.33
MCD	0.60	38.0	7.40 ± 1.54	160.09	10.0[a]	3.39 ± 0.10	1,511.32
MCD	0.65	41.4	8.02 ± 1.09	146.70	0.0	3.24 ± 0.08	1,516.38
MCD	0.70	44.0	7.95 ± 1.03	127.07	0.0	3.13 ± 0.07	1,346.98
MCD	0.75	47.5	7.35 ± 0.94	78.40	0.0	3.00 ± 0.06	793.52

MCD	0.80	52.1	6.45 ± 0.56	46.19	0.0	2.55 ± 0.16	353.99
MCD	0.85	50.8	5.59 ± 0.87	36.86	83.3[a]	2.48 ± 0.07	312.49
MCD	0.90	51.9	5.58 ± 0.56	79.57	93.3[a]	2.41 ± 0.07	268.89
MCD	0.95	53.8	4.56 ± 0.17	—	100.0[a]	2.08 ± 0.10	192.41
Ledoit–Wolf	N/A	62.0	—	8.96	0.0	1.12 ± 0.02	79.23
Full Sample	—	62.0	—	36.65	0.0	0.05 ± 0.01	94.82

Note. MCD = minimum covariance determinant; DOCS = Dimensional Obsessive–Compulsive Scale. Values are means across 30 random seeds. Support fraction is the raw support fraction MCD is instructed to retain, and the retained column gives the support remaining after reweighting. The full-sample row is the untrimmed sample covariance rather than an MCD setting, so it has no support fraction. Restriction ratio is the ratio of mean DOCS scores in excluded versus retained control participants, and is undefined (—) for estimators that retain the full sample. Condition numbers are computed over invertible fits only, and are omitted (—) at support

fractions where no seed produced an invertible matrix. Frobenius distance is measured against the per-seed benchmark covariance. Median D^2 is the median squared Mahalanobis distance of the 263 OCD participants; in the MCD rows it is computed about MCD's own robust location, and in the Ledoit–Wolf and full-sample rows about the reference-sample mean. At every support fraction, the percentage of seeds yielding a feature with zero variance within the retained support set matched the percentage of singular fits exactly, and the degenerate feature was always the Snaith–Hamilton Pleasure Scale. Because the benchmark is generated by resampling these same 62 control participants, the untrimmed row recovers it; the informative untrimmed comparison appears in Table S3.

[a]Where the fitted covariance was singular, squared distances were computed via the Moore–Penrose pseudoinverse, which zeroes the degenerate direction, so median D^2 in those rows reflects ten of the eleven phenotypic dimensions.

Table S3

Synthetic Healthy Control Corroboration (N = 2,000)

Estimator	Support Fraction	Retained (Mean)	Restrict Ratio (Mean ± *SD*)	Condition No. (Mean)	% Singular	Frobenius Dist. (Mean ± *SD*)	Median D^2
MCD	0.50	1,006.8	7.84 ± 1.97	857.45	0.0	3.73 ± 0.06	2,415.85
MCD	0.55	1,107.0	7.88 ± 1.76	937.39	0.0	3.58 ± 0.08	2,157.79
MCD	0.60	1,204.4	8.15 ± 1.95	975.75	0.0	3.43 ± 0.11	2,044.30
MCD	0.65	1,306.5	9.32 ± 1.64	1,003.22	0.0	3.27 ± 0.07	2,187.22
MCD	0.70	1,403.7	8.63 ± 0.90	1,033.71	0.0	3.17 ± 0.05	1,827.88
MCD	0.75	1,488.2	8.77 ± 1.21	1,072.31	0.0	3.07 ± 0.08	1,477.68

MCD	0.80	1,590.2	6.01 ± 0.85	1,148.77	0.0	2.63 ± 0.19	677.34
MCD	0.85	1,665.3	5.93 ± 0.72	1,206.56	0.0	2.42 ± 0.06	510.20
MCD	0.90	1,721.9	4.89 ± 1.08	1,171.92	0.0	2.23 ± 0.18	354.93
MCD	0.95	1,742.7	5.15 ± 1.05	608.62	0.0	2.24 ± 0.19	312.46
MCD	1.00	1,831.7	6.07 ± 1.09	32.92	0.0	2.14 ± 0.20	210.50
Ledoit–Wolf	N/A	2,000.0	—	33.86	0.0	0.33 ± 0.11	94.40
Full Sample	—	2,000.0	—	36.41	0.0	0.33 ± 0.11	95.60

Note. MCD = minimum covariance determinant; DOCS = Dimensional Obsessive–Compulsive Scale. Column definitions follow Table S2; as there, median D^2 is computed about MCD's own robust location in the MCD rows and about the reference-sample mean in the others. Synthetic control samples were generated by the bootstrap procedure described in the Study 2 Method, and within each seed, the synthetic reference sample and the benchmark covariance were drawn independently from the real control sample, so the

sample-covariance and Ledoit–Wolf rows are informative in a way their Table S2 counterparts are not. At this sample size the two are indistinguishable (Frobenius error 0.33 for both), whereas MCD's error ranged from 2.14 at a support fraction of 1.00 to 3.73 at 0.50, roughly 6 to 11 times larger. Shrinkage therefore costs little relative to the untrimmed estimator, whereas MCD remains several times further from the benchmark at every trimming level, even where no trimming is requested. No fit was singular in this setting, because the synthetic generator adds continuous jitter to every value and so cannot produce the exact ties that drive rank deficiency in the real sample.

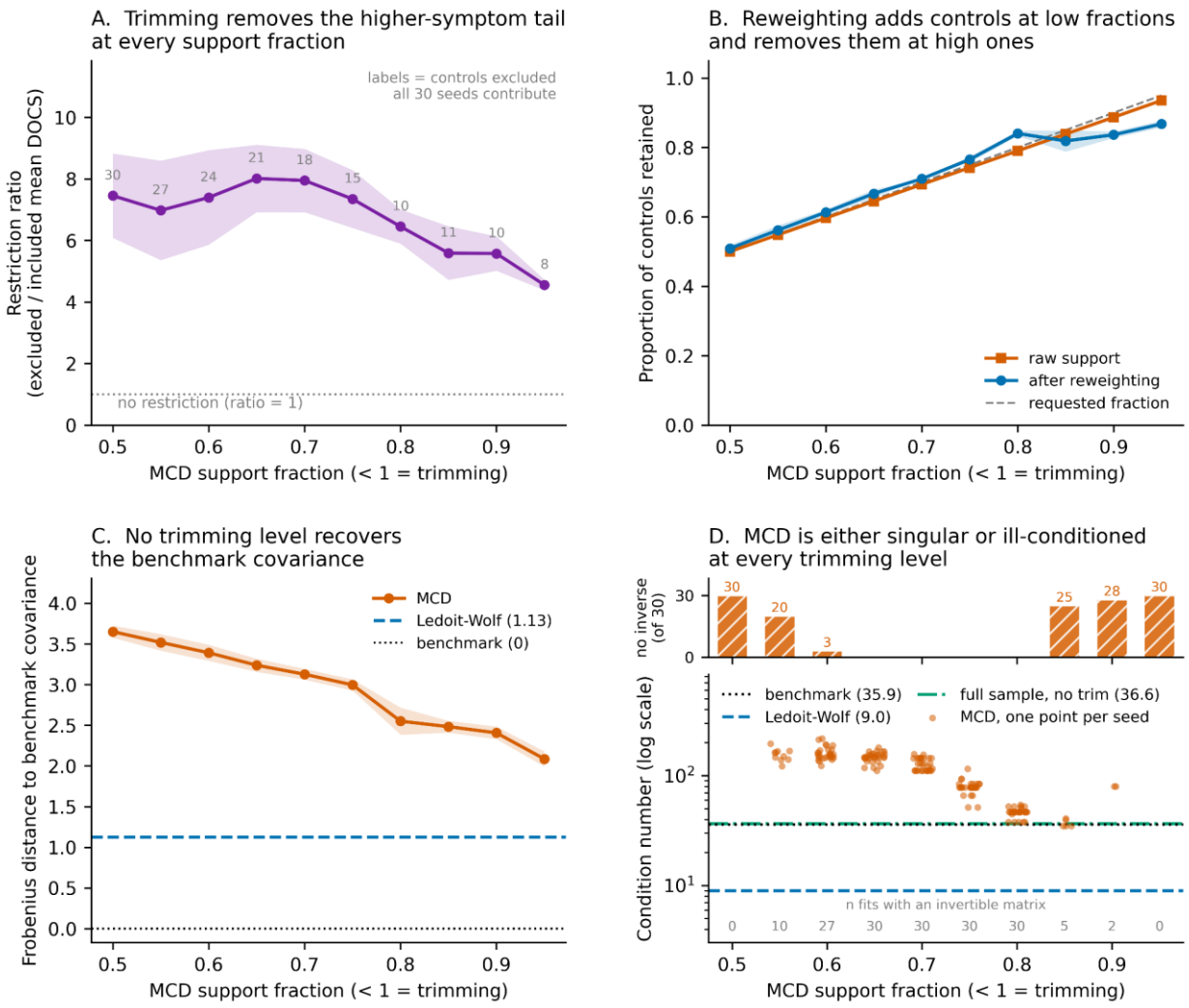


Figure S2. *Estimator Behavior Across the Support-Fraction Grid.* MCD fitted to the real healthy-control sample ($N = 62$) at support fractions from 0.50 to 0.95, across 30 random seeds. (A) Ratio of mean DOCS scores in excluded versus retained control participants. The line is the across-seed mean and the band ±1 *SD*, with the mean number of participants excluded labeled at each point. (B) Proportion of control participants retained before and after MCD's reweighting step, plotted against the requested fraction. (C) Frobenius distance from the fitted covariance to the per-seed benchmark covariance. The untrimmed sample covariance is omitted here because the benchmark is generated by resampling these same 62 participants, which makes that

comparison near-definitional; the corresponding contrast appears in Table S3. (D) Upper panel, the number of seeds at each support fraction for which MCD returned no invertible matrix. Lower panel, condition numbers of the fits that did invert, one point per seed with horizontal jitter, and the count of invertible fits printed beneath each column. The benchmark (35.9) and untrimmed (36.65) reference lines nearly coincide. Panels A through C use all 30 seeds at every support fraction. The lower panel of D uses only the invertible fits, and singular fits are omitted rather than plotted at a nominal value.

Supplement S6: Reference Sample Size Sensitivity

To test whether MCD's covariance estimation error was merely a small-sample artifact that a larger normative reference would resolve, we ran a simulation across HC sample sizes from $N = 62$ to $N = 10{,}000$, repeated over 30 random seeds. Figure S3 and Table S4 present the results.

Synthetic reference samples were generated from the 62 healthy controls by resampling rows with replacement and adding independent Gaussian noise to each feature, with a standard deviation of 0.05 times that feature's sample standard deviation in the real control data. Jitter defines a smoothed empirical reference distribution and removes the exact ties that bootstrap resampling of 62 rows would otherwise produce. The smoothed distribution, not the empirical one, is what the analysis evaluates, so a sensitivity check reports what changes without it. Every sample was transformed by a standardizing scaler fit on the real controls alone, so that all sizes are expressed in the same units and differences across sizes reflect estimation rather than rescaling.

Reference samples were drawn at $n = 62$, 100, 200, 500, 1,000, 2,000, 5,000, and 10,000, over 30 seeds (42 through 71), giving 240 reference samples per estimator. Within each seed, a

benchmark was constructed by the same generator at 100,000 observations, and its mean and covariance define the target the estimators are compared against. Because the benchmark is redrawn for every seed, the comparison is not conditioned on a single benchmark draw. Also within each seed, a deviant group of 263 observations was generated by applying the same resample-and-jitter procedure to the 263 OCD participants after feature imputation, and was held constant across the eight reference sizes, so that any change in its deviation scores comes from the reference-side estimate alone.

The sample covariance, Ledoit–Wolf shrinkage, and MCD at its default support fraction were fitted to each reference sample. Squared distances were computed about a common center, the reference sample mean, so that estimator differences reflect scatter rather than location, and MCD's own robust-location distances were recorded separately. Each fit contributes Frobenius and relative Frobenius error against the benchmark covariance, the corresponding error for the precision matrix, numerical rank and condition number, and distance-level outcomes against the benchmark distances, namely median relative error, Spearman agreement, and top-decile overlap. A jitter-sensitivity check repeated the procedure at jitter values of 0, .01, .05, and .10 at sizes 62, 1,000, and 10,000.

Figure S3 and Table S4 give the results. The sample covariance and Ledoit–Wolf both converged on the benchmark as the reference sample grew, with Frobenius error falling from 2.07 and 1.86 at $n = 62$ to 0.16 for both at $n = 10{,}000$, and relative precision-matrix error from 2.69 and 0.55 to 0.03. MCD did not converge. Its Frobenius error was 3.46 at $n = 62$ and 3.73 at $n = 10{,}000$, and its precision-matrix error fell only from 58.7 to 36.3, remaining three orders of magnitude above the two alternatives.

The consequences for the deviant group followed the same pattern. Against a benchmark median of 93.6, Ledoit–Wolf assigned 94.6 at $n = 62$ and 94.1 at $n = 10{,}000$, the sample covariance 142.7 falling to 94.4, and MCD between 1,644 and 2,948 at every size. Rank agreement with the benchmark distances reached .999 for both full-sample estimators by $n = 5{,}000$ while MCD moved from .704 to .764, and top-decile overlap rose from .48 and .61 to .97 for the two alternatives while MCD stayed between .25 and .28 throughout. Increasing the reference sample improved recovery for both full-sample estimators, while substantial covariance and distance distortion persisted under MCD.

The instability was as marked across runs as across sample sizes. At $n = 62$, MCD's per-seed medians for the deviant group had a median of 2,538 with quartiles at 1,906 and 3,169, against a median of 94 with quartiles at 73 and 103 for Ledoit–Wolf. The same separation held at every reference size. At every sample size and in every seed, the controls MCD excluded had a higher mean DOCS score than those it retained.

The jitter check shows what smoothing changes. Without jitter, MCD's fitted covariance was rank-deficient in 93.3% of seeds at $n = 62$, 96.7% at $n = 1{,}000$, and 100% at $n = 10{,}000$. No rank-deficient fits occurred under any positive-jitter condition, and the covariance-estimation error was essentially unchanged: at $n = 62$ it was 3.48 without jitter and 3.46 to 3.49 across jitter values of .01 to .10, with the same pattern at the two larger sizes.

A seed-count stability check, FigS_seed_convergence in the repository, plots the running mean of MCD's Frobenius error at $n = 1{,}000$. For that quantity the running estimate changed little after 20 seeds.

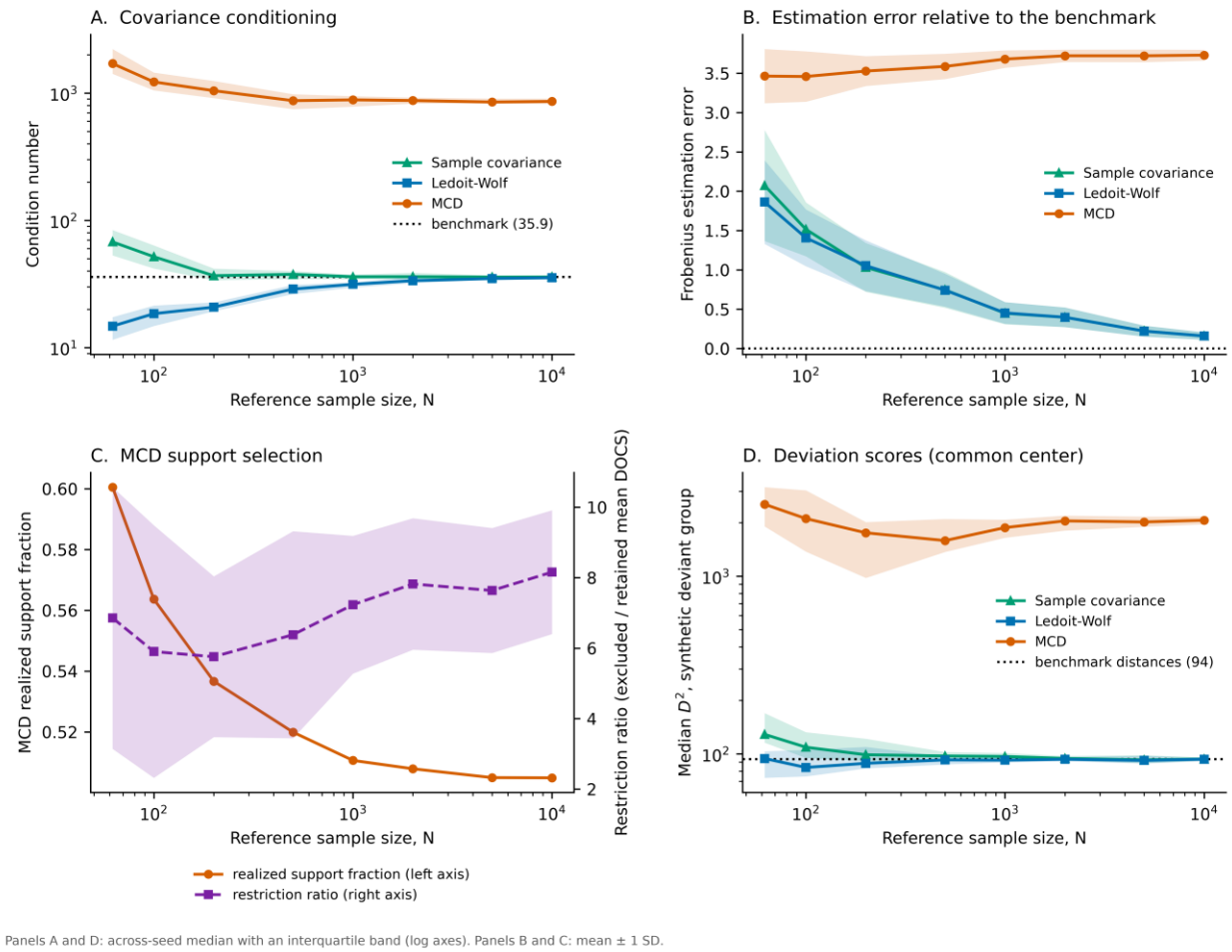


Figure S3. *Estimator behavior across reference sample size.* Three estimators were fitted to synthetic healthy-control samples at $N = 62$ to 10,000, generated by bootstrap resampling with Gaussian jitter from the real control data, across 30 random seeds. A synthetic deviant group ($n = 263$) was held constant across sample sizes within each seed, so variation reflects the reference-side covariance estimate. (A) Condition number of the fitted covariance, with the benchmark covariance's own value marked at 35.9. (B) Frobenius distance from the fitted covariance to the per-seed benchmark. (C) MCD's realized support fraction after reweighting, on the left axis, and the ratio of mean DOCS scores in excluded versus retained reference participants, on the right; neither is defined for the full-sample estimators. (D) Median squared Mahalanobis distance assigned to the deviant group, with the benchmark distances marked at 94. Panels A and D give

the across-seed median with an interquartile band on logarithmic axes; panels B and C give the mean ± 1 SD. MCD = minimum covariance determinant; DOCS = Dimensional Obsessive–Compulsive Scale.

Table S4

Simulation Summary

	Frobenius error			Precision-matrix error			Median D^2			
n	**Sample**	**LW**	**MCD**	**Sample**	**LW**	**MCD**	**Sample**	**LW**	**MCD**	**Benchmark D^2**
62	2.07	1.86	3.46	2.69	0.55	58.69	142.7	94.6	2,947.8	93.6
100	1.52	1.41	3.46	0.71	0.48	52.08	116.1	91.8	2,233.9	93.6
200	1.03	1.05	3.53	0.26	0.33	43.14	105.9	94.8	1,722.5	93.6
500	0.75	0.74	3.59	0.15	0.19	36.85	97.3	92.3	1,644.4	93.6
1,000	0.45	0.45	3.68	0.09	0.12	37.21	95.7	92.6	1,897.8	93.6
2,000	0.40	0.40	3.72	0.06	0.07	36.30	95.2	94.0	1,996.7	93.6
5,000	0.22	0.22	3.72	0.04	0.04	36.57	93.6	93.0	2,030.7	93.6
10,000	0.16	0.16	3.73	0.03	0.03	36.26	94.4	94.1	2,074.5	93.6

Note. Sample = sample covariance; LW = Ledoit–Wolf; MCD = minimum covariance determinant. Frobenius and precision-matrix errors are means across the 30 seeds. Median D^2 is the median across the 263 deviant observations within a seed, averaged over seeds; that group is held constant across sample sizes. Frobenius distance is measured against the per-seed benchmark covariance, whose own condition number was approximately 35.9; smaller values indicate closer recovery. Precision-matrix error is the relative Frobenius error of the inverse, which is the quantity the distance calculation uses. Realized support fraction and restriction ratio are plotted in Figure S3C rather than tabulated here.

Supplement S7: Deviation Score Distributions

Figure S4 gives the distribution of multidimensional deviation scores in the healthy-control and OCD groups under the retained Ledoit–Wolf covariance, on a logarithmic scale.

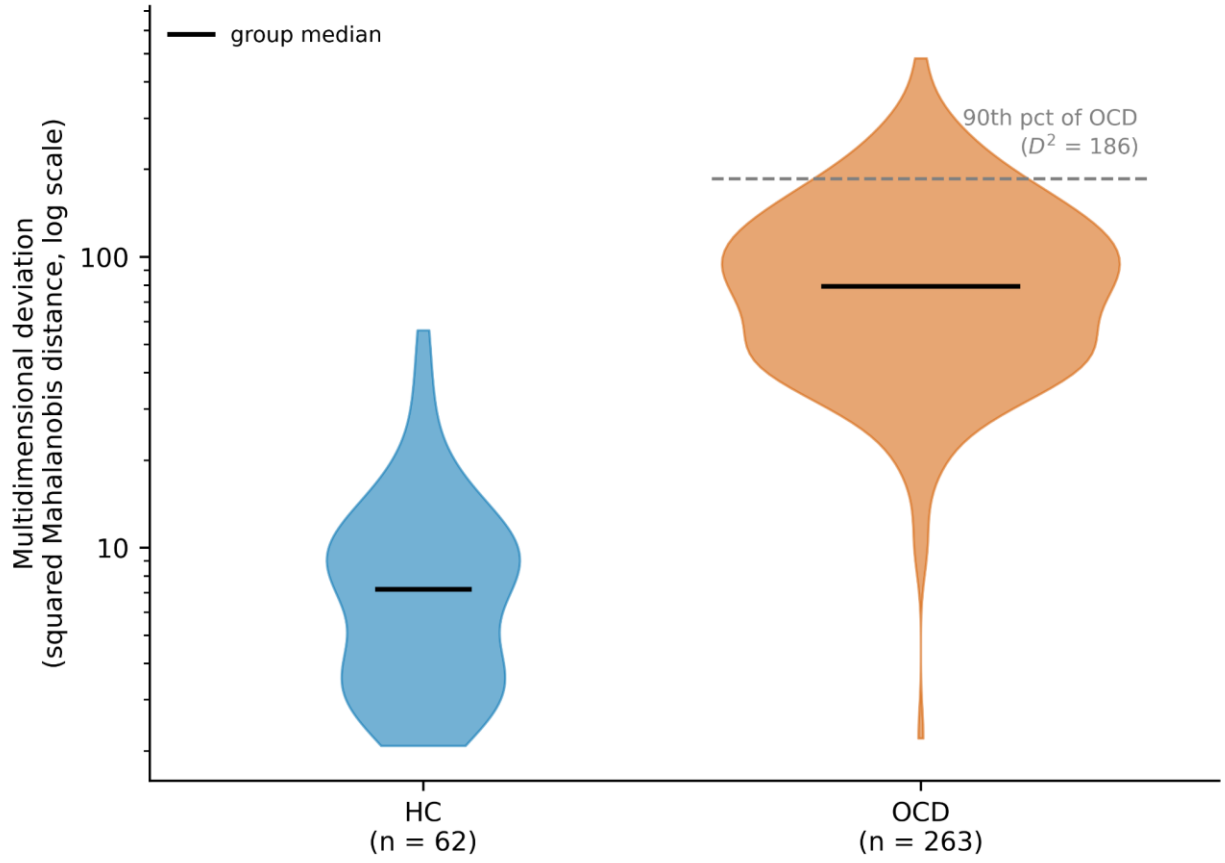


Figure S4. *Deviation scores in healthy controls and in OCD.* Violin plots show multidimensional deviation, computed as the squared Mahalanobis distance (D^2) from the Ledoit–Wolf reference covariance fitted to the full healthy-control sample, on a logarithmic scale. HC (n = 62); OCD (n = 263). Kernel densities were estimated on the log scale. Horizontal lines mark group medians (HC = 7.22, OCD = 79.23). The dashed line marks the 90th percentile of the OCD distribution (D^2 = 185.65), which defines the top-decile subgroup characterized in Table S5. Violin width is scaled by the square root of group size.

Supplement S8: Directional Profiling of the Highest-Deviation Patients

Deviation is defined for every patient regardless of its incremental predictive value, and the phenotypic composition of the most deviant patients is descriptively informative. Taking the 90th percentile of the deviation distribution over the full OCD cohort (D^2 = 185.65) identifies 27

patients. For each of the eleven features, a *z* score was computed relative to the Ledoit–Wolf normative location and scale, and Table S5 reports the mean profile.

The largest displacements were on the symptom dimensions, with obsessive–compulsive severity and general depression exceeding nine standard deviations above the normative mean and harm avoidance and somatic burden above six. These magnitudes partly reflect the narrow spread of these measures in screened controls. Against a low-variance baseline, modest raw elevations map to large standardized distances. This is the same structure that drives the estimator comparison, since the few controls scoring above the floor carry most of the variance on those dimensions and are the observations MCD's trimming removed.

The personality pattern was subtler. Neuroticism was elevated while extraversion and conscientiousness were reduced, with agreeableness and openness near normative levels. The most geometrically deviant patients are thus defined primarily by severe, broad-spectrum symptom elevation accompanied by a milder but distinct personality signature. Two limits apply to this profile. The subgroup is defined by the deviation score itself, so its elevation on the features that drive that score is partly definitional, and the profile was not related to any outcome. It is reported to show what the resulting scores look like rather than to argue for their clinical use.

Table S5

Mean z*-Score Profile of the Top-Decile Deviant OCD Subgroup*

Phenotypic dimension	**Mean *z***
DOCS total	+9.56
gBDI	+9.33
OCTCDQ harm avoidance	+6.50
Somatic symptom burden (PSC)	+6.30
OCTCDQ incompleteness	+4.64
SHAPS anhedonia	+3.90
Neuroticism	+2.58
Agreeableness	−0.49
Openness/Intellect	−0.52

Conscientiousness	−1.31
Extraversion	−1.80

Note. $n = 27$. Values are mean per-feature z-scores for the top decile of the OCD deviation distribution (90th-percentile cutoff $D^2 = 185.65$), expressed relative to the Ledoit–Wolf normative location and scale. DOCS = Dimensional Obsessive–Compulsive Scale; gBDI = General Depression subscale of the Beck Depression Inventory–II; OCTCDQ = Obsessive–Compulsive Trait Core Dimensions Questionnaire; PSC = Physical Symptom Checklist; SHAPS = Snaith–Hamilton Pleasure Scale.

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